Xiaochen Ma

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Research Interests

LLM data infrastructure and distributed data processing systems, with a focus on lineage-aware dataflows, operator management, and scalable data preparation with Ray and Apache Spark.

Education

The Hong Kong University of Science and Technology (HKUST) *Hong Kong SAR, China*
Ph.D. Candidate in Computer Science and Engineering *Feb. 2026 – Expected 2028*
Advisors: Prof. Binhang Yuan and Prof. Wentao Zhang.

Sichuan University *Chengdu, China*
B.Eng. in Computer Science and Technology *Sep. 2019 – Jun. 2023*
Outstanding Undergraduate Thesis, Sichuan University.

Research and Industry Experience

Tencent Hunyuan, Foundation Model Department *Beijing, China*
Research Intern, Qingyun Program *Jun. 2026 – Present*

- Building data infrastructure for large-scale LLM data preparation on Ray and Apache Spark, with an emphasis on usable, distributable, and reproducible operator management.
- Developing distributed pipelines for data deduplication, clustering, and large-scale document and video processing.

Peking University *Beijing, China*
Research Assistant, advised by Prof. Wentao Zhang *Jan. 2025 – Jan. 2026*

- Led the development of DataFlow, coordinating a team of more than 10 contributors across system design, implementation, code review, integration, and releases.
- Developed data-preparation systems for LLMs, including pipeline and operator abstractions, storage and serving interfaces, and Ray-based distributed execution.

Sichuan University *Chengdu, China*
Full-time Research Assistant, advised by Prof. Jizhe Zhou *Jul. 2023 – Jun. 2024*

- Co-led the design and implementation of IMDLBenCo, a modular benchmark and codebase for image manipulation detection and localization.
- Conducted research on image forensics, reproducible evaluation, and efficient vision architectures, resulting in publications at NeurIPS, AAAI, and ICCV.

Selected Systems and Open-Source Projects

DataFlow [Code] [Paper] *Project Lead*

- A unified framework for LLM data preparation with nearly 200 reusable operators and six domain-general pipelines spanning generation, cleaning, evaluation, and orchestration; accepted for publication at **SIGMOD'27**.
- Helped define and deliver the core architecture while coordinating a 10+ member team; contributed to pipeline compilation and execution, the operator registry, storage, serving, CLI tooling, and distributed execution.

RayOrch [Code] [Paper] *First Author and Project Initiator*

- Designed a programming model and distributed execution engine for foundation-model data preparation, where one input can expand into many records while preserving parent-child lineage, ordering, completion, and result assembly.
- Developed compiler and runtime mechanisms for ordered expansion, parent-scoped failures, and Ray-based execution across large-scale document and video workloads.

Flash-MinerU [Code] *Initiator and Core Developer*

- Developed a scalable Ray-based acceleration layer for MinerU PDF-to-Markdown processing, refactoring sequential stages into an asynchronous multi-GPU and multi-node pipeline.

Selected Publications

* *Equal contribution. Full list: Google Scholar.*

1. **Xiaochen Ma**, Zimo Meng, Junzhu Liang, et al. *RayOrch: Programming and Executing Lineage-Controlled Multi-Grain Dataflows for Foundation-Model Data Preparation.* **arXiv'26**. [Paper] [Code]
2. Hao Liang*, **Xiaochen Ma***, et al. *DataFlow: An LLM-Driven Framework for Unified Data Preparation and Workflow Automation in the Era of Data-Centric AI.* Accepted for publication at **SIGMOD'27**. [Paper] [Code]
3. **Xiaochen Ma***, Xuekang Zhu*, Lei Su*, Bo Du*, Zhuohang Jiang*, Bingkui Tong*, Zeyu Lei*, Xinyu Yang*, Chi-Man Pun, Jiancheng Lv, and Jizhe Zhou. *IMDL-BenCo: A Comprehensive Benchmark and Codebase for Image Manipulation Detection and Localization.* **NeurIPS'24 Spotlight**. [Paper] [Code]
4. Xuekang Zhu*, **Xiaochen Ma***, Lei Su, Zhuohang Jiang, Bo Du, Xiwen Wang, Zeyu Lei, Wentao Feng, Chi-Man Pun, and Jizhe Zhou. *Mesoscopic Insights: Orchestrating Multi-scale and Hybrid Architecture for Image Manipulation Localization.* **AAAI'25**. [Paper] [Code]
5. Bo Du*, Xuekang Zhu*, **Xiaochen Ma***, Chenfan Qu*, Kaiwen Feng*, Zhe Yang, Chi-Man Pun, Jian Liu, and Ji-Zhe Zhou. *ForensicHub: A Unified Benchmark and Codebase for All-Domain Fake Image Detection and Localization.* **NeurIPS'25**. [Paper] [Code]
6. Lu Ma, Hao Liang, Meiyi Qiang, Lexiang Tang, **Xiaochen Ma**, et al. *Learning What Reinforcement Learning Can't: Interleaved Online Fine-Tuning for Hardest Questions.* **ICLR'26**. [Paper]
7. Yuran Wang, Bohan Zeng, Chengzhuo Tong, Wenxuan Liu, Yang Shi, **Xiaochen Ma**, Hao Liang, Yuanxing Zhang, and Wentao Zhang. *Scone: Bridging Composition and Distinction in Subject-Driven Image Generation via Unified Understanding-Generation Modeling.* **CVPR'26 Highlight**. [Paper]
8. Bo Du, **Xiaochen Ma**, Xuekang Zhu, et al. *Can We Build a Monolithic Model for Fake Image Detection? SICA: Semantic-Induced Constrained Adaptation for Unified-Yet-Discriminative Artifact Feature Space Reconstruction.* **ICML'26**. [Paper] [Code]
9. Jizhe Zhou, **Xiaochen Ma**, Xia Du, Ahmed Y. Alhammadi, and Wentao Feng. *Pre-training-free Image Manipulation Localization through Non-Mutually Exclusive Contrastive Learning.* **ICCV'23**. [Paper]

Academic Service

Reviewer: AAAI'27; NeurIPS'26; BMVC'26; ECCV'26; CVPR'26; ICLR'26; AISTATS'26; AAAI'25; NeurIPS'25; ICCV'25; ICML'25; CVPR'25; ICLR'25; AISTATS'25; NeurIPS'24; ACM MM'24.

Honors and Awards

Outstanding Undergraduate Thesis, Sichuan University *Jun. 2023*
National Second Prize, China Software Cup College Student Software Design Competition *Oct. 2022*

Technical Skills and Languages

- **Programming and systems**: Python, C/C++, Ray, Apache Spark, PyTorch, Docker, Linux, Git, Bash, LaTeX.
- **Languages**: Mandarin (native); English (IELTS 7.0, GRE 324).